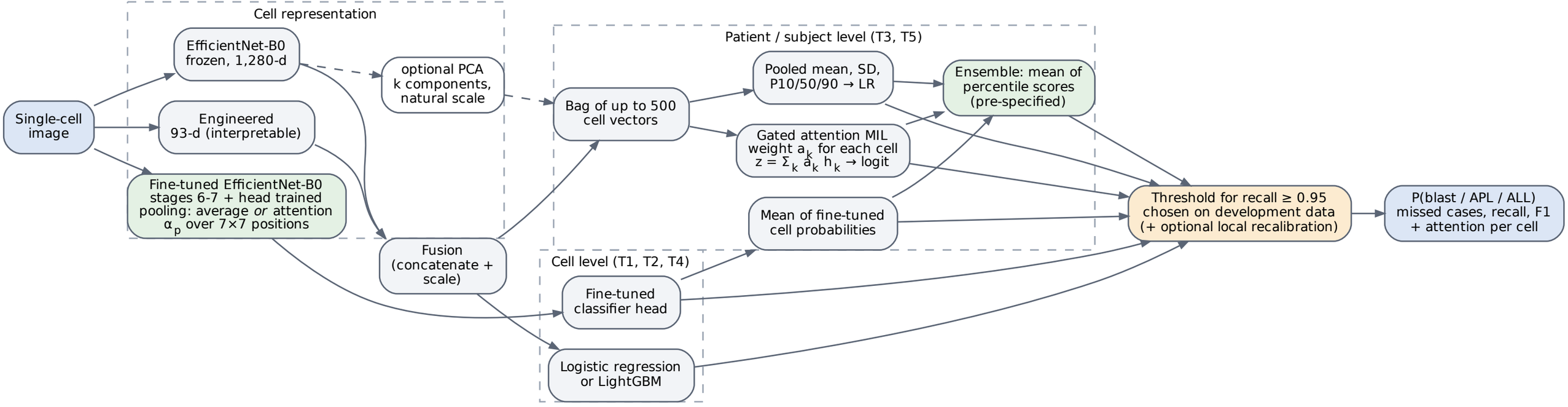
Cell representation
Patient / subject level (T3, T5)
Cell level (T1, T2, T4)
Single-cell image
EfficientNet-B0 frozen, 1,280-d
Engineered 93-d (interpretable)
Fine-tuned EfficientNet-B0 stages 6-7 + head trained pooling: average or attention α_p over 7×7 positions
optional PCA k components, natural scale
Fusion (concatenate + scale)
Bag of up to 500 cell vectors
Pooled mean, SD, P10/50/90 → LR
Gated attention MIL weight a_k for each cell $z = \Sigma_k a_k h_k$ → logit
Mean of fine-tuned cell probabilities
Ensemble: mean of percentile scores (pre-specified)
Fine-tuned classifier head
Logistic regression or LightGBM
Threshold for recall ≥ 0.95 chosen on development data (+ optional local recalibration)
P(blast / APL / ALL) missed cases, recall, F1 + attention per cell